carefully we see that events that occur at two separate places at the same time, as seen by Moe in $S'$, do *not* happen at the same time as viewed by Joe in $S$. If one event occurs at point $x_1$ at time $t_0$ and the other event at $x_2$ and $t_0$ (the same time), we find that the two corresponding times $t'_1$ and $t'_2$ differ by an amount

$$t'_2 - t'_1 = \frac{u(x_1 - x_2)/c^2}{\sqrt{1 - u^2/c^2}}.$$

This circumstance is called "failure of simultaneity at a distance," and to make the idea a little clearer let us consider the following experiment.

Suppose that a man moving in a spaceship (system $S'$) has placed a clock at each end of the ship and is interested in making sure that the two clocks are in synchronism. How can the clocks be synchronized? There are many ways. One way, involving very little calculation, would be first to locate exactly the midpoint between the clocks. Then from this station we send out a light signal which will go both ways at the same speed and will arrive at both clocks, clearly, at the same time. This simultaneous arrival of the signals can be used to synchronize the clocks. Let us then suppose that the man in $S'$ synchronizes his clocks by this particular method. Let us see whether an observer in system $S$ would agree that the two clocks are synchronous. The man in $S'$ has a right to believe they are, because he does not know that he is moving. But the man in $S$ reasons that since the ship is moving forward, the clock in the front end was running away from the light signal, hence the light had to go more than halfway in order to catch up; the rear clock, however, was advancing to meet the light signal, so this distance was shorter. Therefore the signal reached the rear clock first, although the man in $S'$ thought that the signals arrived simultaneously. We thus see that when a man in a spaceship thinks the times at two locations are simultaneous, equal values of $t'$ in his coordinate system must correspond to *different* values of $t$ in the other coordinate system!